W. Matthew Alampay Davis

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Department of Economics
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Education

Columbia University	University of Oxford	New York University
Dept. of Economics, GSAS	Nuffield College	Stern School of Business
Ph.D Economics (2025)	*M.Phil* Economics (2019)	*B.Sc* Business & Political Economy (2015)

Research

Development economics · Environmental/Climate Economics · Political Economy

Job market paper

Climate inequality

This paper documents a new result—that transitory temperature shocks tend to increase within-country income inequality substantially and persistently—and considers its implications for the welfare economics of climate change. First, I use grouped income data from distributional national accounts to estimate dynamic growth incidence responses to an identified pulse of temperature over a multidecadal projection horizon. This is accomplished by refining newly developed time series methods to flexibly address several commonly raised statistical concerns with minimally restrictive parametric assumptions. Results depict steep non-linear income gradients funneling marginal costs to lower-income groups and concentrating marginal relative benefits in narrow high-income quantiles. This pattern of compounding inequality is found to be surprisingly consistent across geographies while still reproducing the stark temperature-based divergence reported in abundant empirical work when integrated over observed income distributions. The distributional and aggregate results are reconciled by rejecting speculative but frequently offered bottom-up explanations for the aggregate divergence, demonstrating the first-order importance of departing from standard distribution-agnostic approaches.

I then integrate this dynamic impulse response model with the outputs of a suite of climate models to simulate the influence of anthropogenic climate change on the global income distribution first i) between 1960-2016 and then ii) under a variety of climate forcing scenarios projected through to 2100. I integrate the results from this counterfactual exercise with a stylized model incorporating a variety of alternative social welfare functions from which optimal fiscal policy can be derived. I close by considering these findings in relation to the prevailing utilitarian cost-benefit paradigm in the economics of climate change.

Work in progress

Political transition under environmental stress

Economic disruption, cultural anti-persistence, and intimate-partner violence

with Tanushree Goyal

Elite capture and identity politics: A formal model of substantive representation

Publications

Large potential reduction in economic damages under UN mitigation targets

with Marshall Burke, Noah S. Diffenbaugh — *Nature* (2018)

We present a probabilistic framework for assessing aggregate economic impacts of anthropogenic warming. Our construction decomposes uncertainty associated with mid-century and end-of-century GDP projections into error associated with i) econometric estimation of the economic effects of environmental change, ii) climate models of the spatial distribution of anthropogenic warming, iii) the projected schedule of greenhouse gas concentrations associated with radiative forcings, and iv) the social discounting regime of choice. We apply this framework to characterize the economic benefits of climate policy, emphasizing how achieving the most ambitious mitigation targets of the 2015 Paris Agreement would obviate essentially certain economic calamity concentrated in developing countries.

Combining satellite imagery and machine learning to predict poverty

with Neal Jean, Marshall Burke, Michael Xie, David B. Lobell, Stefano Ermon — *Science* (2016)

We demonstrate a scalable method to address the scarcity and prohibitive expense of collecting disaggregated welfare data in the developing world by extracting economic information from an unconventional but abundant data source with increasingly frequent and near-global coverage: daytime satellite photography. Our "transfer learning" pipeline proceeds by first assigning a convolutional neural network model pre-trained for generic image classification the task of identifying features in georeferenced satellite images predictive of night-time luminosity, a crude proxy for economic activity. In effect, the CNN learns to produce a nonlinear mapping from high-resolution unstructured image data to low-dimensional vector representations of their most economically relevant features. Ridge regression models are then trained on survey data to refine the feature selection for out-of-sample predictive performance. In an exploratory application to five diverse sub-Saharan African countries—Nigeria, Tanzania, Uganda, Malawi, and Rwanda—our open-source models trained on publicly available data are found to explain up to 75% of the variation in village-level consumption and asset wealth measured by traditional household surveys.

Referee service — *Journal of Development Economics*

Teaching

Graduate Teaching Fellow at Columbia University (nine semesters, 2020–2024)
Research consultant to all UG/MA economics thesis writers and research assistants
ECON-4999 – Senior honors thesis workshops (design and instruction)
ECON-3412 – Introduction to econometrics (TA)
ECON-3211 – Intermediate microeconomics (TA)
ECON-2257 – The global economy (TA)

Past affiliations

Research associate at Climate Econometrics (2017–2019)
Nuffield College, University of Oxford
supervised by David F. Hendry, Elizabeth Baldwin

Pre-doctoral scholar at Center on Food Security and the Environment (2015–2017)
Dept. of Earth System Science, Stanford University
supervised by Marshall Burke, David Lobell

Honors

Program for Economic Research Summer Fellowship (2024)
Columbia economics department award supporting summer research activities

Wueller Fourth-Year Pre-Dissertation Award (2023)
Columbia economics department runner-up best dissertation proposal

UC Berkeley and Sloan Foundation Diversity Fellowship (2022)
Awarded to five doctoral students in environmental and energy economics

Columbia GSAS Dean's Fellowship and Dissertation Fellowship (2019)
Full-tuition scholarship and living stipend for up to six years of graduate study at Columbia

Oxford M.Phil top-scoring economics thesis (2019)
Department's first 'exceptional' (mark of 80+) thesis in at least four years

Nuffield College–Department of Economics Joint Studentship (2017)
Full-tuition scholarship and living stipend for up to four years of graduate study at Oxford

References

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Personal

Background	Philippines (nationality) · Indonesia (13 years) · Vietnam
Study and career	United States · United Kingdom · China
Immediate family	Philippines · Australia · Canada